Larkhall NDE Services Inc. **Inspection Report**

Site: Hamilton Plant

No.	Status	Date
LNDE-RPT-20150427-019-R0	Issued	2015-05-08

Shear-Wave UT of Circumferential and Attachment Welds - Lean Amine

Scope and Objectives

This report details the external shear-wave ultrasonic testing (UT) performed on the circumferential and attachment welds for lines 4-LA-101, 4-LA-102, and 4-LA-104 within the ATU-100 at the Hamilton Plant. The inspection was initiated following the MOC requirements outlined in QSCP-DMR-2014-002, which addressed the DEA strength increase and the associated impact on corrosion rates and inspection intervals for amine circuits.

The primary objective was to assess the integrity of the weldments in lean amine service, specifically targeting potential cracking or volumetric flaws. Line 4-LA-104, serving the lean side of the L/R exchanger, operates under a design pressure of 2758 kPa(g) and a design temperature of 138 °C. Lines 4-LA-101 and 4-LA-102, both lean amine supply lines, are rated for 2413 kPa(g) at 93 °C. All three lines are constructed from A106 Gr.B carbon steel with a nominal wall thickness of 6.02 mm.

Personnel and Responsibilities

The field inspection and data acquisition were conducted by Marcus Chen, acting as the lead UT technician. As an SNT-TC-1A UT Level II, Marcus Chen was responsible for the calibration of the ultrasonic equipment, surface preparation oversight, and the execution of the shear-wave scans in accordance with the site-specific requirements.

Technical oversight and final validation of the inspection results were performed by David Park, the Quality Manager. In the capacity of an SNT-TC-1A UT Level III, David Park ensured that the examination sensitivity and flaw characterization methods adhered to the established quality management system and the technical specifications required for the ATU-100 amine service lines.

Inspection Methodology

The examination was performed under QSCP-NDT-001 Rev 2, which provides the site-specific procedure for ultrasonic measurements and includes provisions for insulation removal and reinstatement. The shear-wave techniques utilized for the circumferential and attachment welds were executed to meet the requirements of ASME BPVC Section V, ensuring adequate coverage of the weld root and heat-affected zones.

Surface preparation involved the removal of insulation at designated inspection ports as mandated by the CUI program requirements in QSCP-NDT-001 Rev 2. The ultrasonic testing focused on identifying any service-induced defects that could compromise the t-min of 3.18 mm. The inspection was conducted from the external pipe surface using calibrated shear-wave transducers to evaluate the full volume of the weldments.

Summary of Results

The shear-wave UT of the circumferential and attachment welds on 4-LA-104, 4-LA-101, and 4-LA-102 revealed no relevant indications. No evidence of cracking, lack of fusion, or significant volumetric inclusions was detected during the scan. All examined areas showed structural integrity consistent with the original fabrication standards and the current service requirements defined in QSCP-DMR-2014-002.

Based on the findings, the lean amine lines remain within the acceptable limits for continued service. As these lines are classified as Class 2 under the site rules for amine service, the next thickness reading and external visual inspection shall be scheduled according to the 10-year and 5-year intervals respectively, unless the remaining life calculations derived from future CML data necessitate an earlier intervention.

Sign-off	Name	Function	Date	Signature
Prepared	Marcus Chen	lead UT technician, SNT-TC-1A UT/MT/PT Level II; CAN/CGSB-48.9712 UT Level II	2015-05-08	_____
Reviewed	David Park	Level III / Quality Manager, SNT-TC-1A UT Level III	2015-05-08	_____